



Lesson 6: Lists and Simple Data



Welcome Back!

Remember Scratch's list variable, where you could keep adding items and check item #1, #2, #3? Python has lists too — and today we learn how to use them!



Think Like a Coder

A list is like a row of connected boxes, all sharing one label, holding many pieces of information in order.



Step 1: Create Your First List

Type and run:

```
crew = ["Nova", "Max", "Zara"]
print(crew[0])
print(crew[1])
print(crew[2])
```



Robot Tip

Lists start counting at **0**, not 1! So `crew[0]` is the FIRST item ("Nova"), not the second.



Debug Detective

This code has a bug. Can you find it?

```
crew = ["Nova", "Max", "Zara"]
print(crew[3])
```

Hint: how many items are in the list, and what are their valid index numbers?

The Fix:

```
crew = ["Nova", "Max", "Zara"]
print(crew[2])
```

The list only has 3 items, so the valid indexes are 0, 1, and 2. Index 3 doesn't exist — that caused an `IndexError` !

Step 2: Loop Through a List

```
crew = ["Nova", "Max", "Zara"]
for member in crew:
    print(f"Crew member: {member}")
```

This prints every single item in the list, one at a time, without us needing to know how many there are!

Step 3: Pick a Random Item

```
import random

crew = ["Nova", "Max", "Zara"]
captain = random.choice(crew)
print(f"Today's captain is {captain}!")
```

Run it a few times — the captain changes randomly each time!

Mini Challenge

Add a 4th crew member and run the loop again to see it appear automatically:

```
crew = ["Nova", "Max", "Zara", "Kai"]
for member in crew:
    print(f"Crew member: {member}")
```

★ Bonus Challenge

Use `len()` to find out how many items are in a list, and combine it with `range()` :

```
crew = ["Nova", "Max", "Zara"]
print(f"There are {len(crew)} crew members.")
for i in range(len(crew)):
    print(f"Position {i}: {crew[i]}")
```

🎨 Make It Yours

Build "The Random Mission Picker" — check [Activities.md](#) for full steps!

🏆 Victory Moment

You just built a program that stores a whole group of information AND picks something random from it — that's how games choose random events and rewards!

🔮 Coming Up Next

In Lesson 7, we'll combine everything we've learned — variables, input, if/else, and more — to build a real **mini quiz app** !